

Homework 4-SSIE 553

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Due - 03/02/2026 by 11:59pm through Brightspace

Question 1[5+5+5pts]: Consider the following Linear Programming Problem and let x_2 and s_1 be the basic variables.

$$\text{Max } z = -x_1 + x_2$$

Subject to:

$$2x_1 + x_2 \leq 4$$

$$x_1 + x_2 \leq 2$$

$$x_1 \geq 0, x_2 \geq 0$$

- i Determine x_B, x_N, B, N, c_B, c_N and b as defined in lecture 9 for the given set of basic variables.
- ii Construct the Simplex Tableau for the given set of basic variables.
- iii Comment on the feasibility and optimality of the Basis defined by the given set of basic variables.

Question 2[10+10+5pts]: Company X can manufacture three types of candy bar. Each candy bar consists of sugar and chocolate. Total of 50oz and 100oz of chocolate are available. After defining x_i to be number of Type i candy bars manufactured, Company X should solve the following LP:

$$\text{Max } z = 3x_1 + 7x_2 + 5x_3 \text{ (Total Profit in cents)} \quad (1)$$

subject to

$$x_1 + x_2 + x_3 \leq 50 \text{ (Sugar Capacity Constraint)} \quad (2)$$

$$2x_1 + 3x_2 + 1x_3 \leq 100 \text{ (Chocolate Capacity Constraint)} \quad (3)$$

$$x_i \geq 0 \quad \forall i = \{1, 2, 3\} \quad (4)$$

After solving the LP, Company X determined the set of optimal basic variables as $\{x_3, x_2\}$

$$B^{-1} = \begin{bmatrix} \frac{3}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

- i For what values of Type 1 candy bar profit does the current basis remain optimal? If the profit for a Type 1 candy bar were 7¢, what would be the new optimal solution?
- ii For what values of Type 2 candy bar profit would the current basis remain optimal? If the profit for a Type 2 candy bar were 19¢, then what would be the new optimal solution?
- iii For what amount of available sugar would the current basis remain optimal? If 60 oz of sugar were available, what would be the optimal profit? How many of each candy bar should the company make?

Question 3[10+5+10+5pts]: Consider the Following LP:

$$\text{Max } z = 60x_1 + 30x_2 + 20x_3 \quad (5)$$

subject to

$$8x_1 + 6x_2 + x_3 \leq 48 \quad (6)$$

$$4x_1 + 2x_2 + 1.5x_3 \leq 20 \quad (7)$$

$$2x_1 + 1.5x_2 + 0.5x_3 \leq 8 \quad (8)$$

$$x_i \geq 0 \quad \forall i = \{1, 2, 3\} \quad (9)$$

where basic variables comprising optimal basis is $\{s_1, x_3, x_1\}$ and

$$B^{-1} = \begin{bmatrix} 1 & 2 & -8 \\ 0 & 2 & -4 \\ 0 & -0.5 & 1.5 \end{bmatrix}$$

- i Determine the range of the objective function coefficient of x_3 such that the current basis remains optimal.
- ii If $c_3^a = 21$, find the new optimal solution.
- iii If $c_3^a = 25$, find the new optimal solution.
- iv Determine the range of the RHS of constraint (6) such that current basis remains optimal.

Question 4[15+15pts]: Solve the following Linear Programming Problem using the Big M and the Two-Phase Simplex Method:

$$\text{Min } z = -4x_1 + 4x_2 + x_3$$

Subject to:

$$x_1 + x_2 + x_3 \leq 2$$

$$2x_1 + x_2 \leq 3$$

$$2x_1 + x_2 + 3x_3 \geq 2$$

$$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0$$